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# Particles of the Universe Meets the Electric Universe

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(revision 2)

**Abstract:** In this paper, the electric nature of the universe is revealed by applying the language of electronics to the energy and forces acting upon subatomic particles.

## Introduction

A quest for grand unification in the physics community is a holy-grail quest in that whomever solves this big problem will likely receive a Nobel Prize. It is generally understood that this unification will involve the unification of relativity and quantum mechanics. That said, the unification with the domain of electronics does not seem to be on anyone's radar, even though, this would also be a huge breakthrough. If all the forces in the universe, including gravity, can be explained in terms of an "electric universe", then the path to unifying relativity and quantum mechanics may be greatly simplified.

The electric universe theory highlights the important role that electricity plays throughout nature and the cosmos. From heartbeats, to neurons in the brain, to the aurora borealis and the electrical discharges we call lightning (not to mention the fact that 99.999% of the universe is in plasma state), "electricity" appears to be everywhere [1,2,3,4,5]. Even in the face of extensive evidence that we live in an electric universe [6,7,8,9], modern physics is still at odds with this concept. Although many still seek to unify quantum mechanics with relativity, the unification of the domain quantum mechanics with the domain of electricity may prove to be more fruitful. If a clear connection between these two domains can be found, then the idea that we live in an electric universe will be very difficult to deny. Such a connection could lead to a better understanding of light and matter and may lead to new technologies that were previously not imaginable. The research presented herein lays the foundation for such a unification.

## 1. Potential of the Universe

In electronics, Ohm's law establishes the relationship between voltage (V), current (I) and resistance (R) as  $V=IR$ . In a typical home, the potential voltage is fixed (e.g. 120 volts in the US and Canada) such that if a load with a known resistance is added to the electrical circuit in the home, electrons flow with a current that satisfies Ohm's law.

If we apply Ohm's law to the universe, we find that it has electrical properties consistent with electronics in the home. The subscript *u* is now applied to Ohm's law for the properties of an electric *universe*.

$$V_u = I_u R_u \quad (1.1)$$

These properties are already known and expressed in constants in electromagnetics. The voltage and resistance properties can be found in the electric constant ( $\epsilon_0$ ) and magnetic constants ( $\mu_0$ ) respectively. In electromagnetics, forces operate at the speed of light ( $c$ ), which can be thought of as the speed of current flow.

$$V_u = \frac{1}{\epsilon_0} \quad (1.2)$$

$$R_u = \mu_0 c \quad (1.3)$$

$$I_u = c \quad (1.4)$$

Substituting above into Ohm's law ( $V=IR$ ), the relationship between the electric and magnetic constants are:

$$\frac{1}{\epsilon_0} = \mu_0 c^2 \quad (1.5)$$

While elegant, there are problems with expressing the electric universe in terms of Ohm's law. One problem with relating these properties is one of units. Specifically, the unit of charge, which is measured in Coulombs [C]. If charge is replaced with wave amplitude [10], which is measured in units of distance [m], then the units of Eqs. 1.3 to 1.5 are now in agreement. In the table below, Coulombs [C] is replaced with meters [m]. Since amperes [A] is Coulombs per second [C/s], it is replaced with the units of velocity [m/s]. In fact, current is a velocity and voltage is now properly expressed as a force.

|              |                   | SI Units                               | Derivation                                                                    | Corrected Units     |
|--------------|-------------------|----------------------------------------|-------------------------------------------------------------------------------|---------------------|
| I            | Current           | $A$                                    | $I_{units} = A = \frac{C}{s} = \frac{m}{s}$                                   | $\frac{m}{s}$       |
| V            | Voltage           | $\frac{kg(m^2)}{s^3(A)}$               | $V_{units} = \frac{kg(m^2)}{s^3(\frac{m}{s})} = \frac{kg(m)}{s^2}$            | $\frac{kg(m)}{s^2}$ |
| R            | Resistance        | $\frac{kg(m^2)}{s^3(A^2)}$             | $R_{units} = \frac{kg(m^2)}{s^3(\frac{m}{s})^2} = \frac{kg}{s}$               | $\frac{kg}{s}$      |
| $\epsilon_0$ | Electric constant | $\frac{A^2 s^4}{kg(m^3)}$              | $\epsilon_{0units} = \frac{(\frac{m}{s})^2 s^4}{kg(m^3)} = \frac{s^2}{kg(m)}$ | $\frac{s^2}{kg(m)}$ |
| $\mu_0$      | Magnetic constant | $\frac{H}{m} = \frac{kg(m)s}{s^2(A)m}$ | $\mu_{0units} = \frac{kg(m)s}{s^2(\frac{m}{s})m} = \frac{kg}{m}$              | $\frac{kg}{m}$      |
| e, q         | Elementary charge | $C$                                    | $e_{units} = C = m$                                                           | $m$                 |

**Table 1** – Corrected units for electrical properties

In Table 1, the elementary charge ( $e$ ) has corrected units of meters. It is the measurement of a single electron [11].

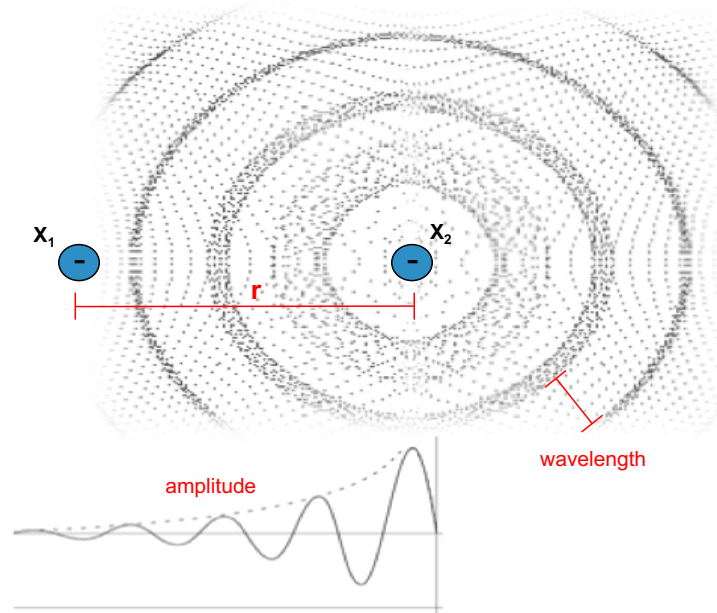
$$e = 1.602 \cdot 10^{-19} [m] \quad (1.6)$$

The collective charge  $q$  is used in the context of this paper as one or more elementary charges within proximity, where constructive or destructive wave interference adds or subtracts to the wave amplitude measured at a given space.

$$q = \Delta e \quad (1.7)$$

What is charge and how can it be the units of distance? A good analogy is sound waves, which are longitudinal waves of moving molecules. Amplitude is the maximum displacement from equilibrium as molecules vibrate, which is measured as a distance. Waves have interference patterns, causing a change in displacement as waves combine. With sound waves, energy is the square of the wave amplitude and its intensity decreases with the square of the distance.

Using these principles from longitudinal waves, imagine measuring a voltage across two points labeled  $X_1$  and  $X_2$ . Each point has a collection of particles, such as electrons ( $-$ ), that contribute wave amplitude. The illustration below shows the displacement (amplitude) of a substance at  $X_2$ , continuing to decrease in amplitude as it spreads out spherically before reaching  $X_1$ .



**Figure 1** – Two points ( $X_1$  and  $X_2$ ) separated at a distance ( $r$ ) where the collective charges at  $X_2$  spread spherically, losing amplitude before reaching the collective charges at  $X_1$ . For illustration purposes, to understand longitudinal wave amplitude, only the effect from charge  $X_2$  is shown in the figure.

According to the Figure 1, at  $X_1$  waves created in close proximity collectively add or subtract wave amplitude, depending on wave phase. Particles in phase will add constructively (Eq. 1.8). For example, two electrons would be two elementary charges ( $2e$ ).

$$X_1 = \Delta e \quad (1.8)$$

The wave interference of the wave traveling in the direction from  $X_2$  to  $X_1$ , is measured at  $X_1$ . It decreases at the square of distance as wave amplitude spreads out spherically, where the distance between  $X_1$  and  $X_2$  is the radius ( $r$ ) of the sphere. For example, three electrons at distance  $r$  would be  $3e/4\pi r^2$ .

$$X_2 = \frac{\Delta e}{4\pi r^2} \quad (1.9)$$

Referring back to Table 1, it is seen that voltage is a force. It is measuring wave energy, across two points in space at distance, where the electrical voltage property of space is fixed ( $V_u$ ). It is wave amplitude that is changing. When there is no displacement of particles, i.e. there is no wave amplitude, there is no force.

$$F = \Delta_u V_u \quad (1.10)$$

$$\Delta_u = X_1 X_2 \quad (1.11)$$

Combining Eqs. 1.8 to 1.11 yields:

$$F = \frac{\Delta e \Delta e}{4\pi r^2} V_u \quad (1.12)$$

Charles-Augustin de Coulomb recognized the constants in this equation and it was given the letter  $k$ , known as Coulomb's constant ( $k$ ). The relationship of  $V_u$  (from Eq. 2), the electric constant and Coulomb's constant is:

$$k = \frac{1}{4\pi\epsilon_0} = \frac{V_u}{4\pi} \quad (1.13)$$

Replacing  $q$  for  $\Delta e$  (Eq. 1.7) and  $k$  for  $V_u/4\pi$  (Eq. 1.13) in Eq. 1.12 yields **Coulomb's law**.

$$F = k \frac{qq}{r^2} \quad (1.14)$$

## 2. @ Electron Classical Radius

In this section, it will be shown that, when voltage is measured across two elementary charges at the electron's classical radius ( $r_e$ ), the electron's energy and mass can be derived. It is not suggested that two elementary charges are physically at this location, but rather, it is a representation of constructive waves that converge on a charged particle, such as the electron, according to Huygen's principle as illustrated below.



a) charge at electron radius

b) distant charges form a wavefront (Huygen's principle)

**Figure 2** – a) the simulated effect of two electrons separated at a distance of the classical electron radius, forming standing waves; a potential representation of the three-dimensional effect of b) distant charges forming a wavefront according to Huygen's principle to achieve the same standing wave. Motion results from unequal wavefront amplitude.

Since force is synonymous with voltage (from Eq. 1.10), using Ohm's law, the following is also true:

$$F = \Delta_u I_u R_u \quad (2.1)$$

Current ( $I_u$ ) and resistance ( $R_u$ ) were established in Eqs. 1.4 and 1.3 respectively. They are substituted into Eq. 2.1 and *rearranged* to become Eq. 2.2. The constructive wave interference variable (from Eqs. 1.11, 1.8 and 1.9) is then substituted to become Eq. 2.3.

$$F = \mu_0 c^2 (\Delta_u) \quad (2.2)$$

$$F = \mu_0 c^2 \left( \frac{\Delta e \Delta e}{4\pi r^2} \right) \quad (2.3)$$

Eq. 2.3 is a force, but a single charged particle is not measured as a force. It is typically measured as a mass or energy. Energy is a force exerted over a distance ( $d$ ), expressed in Eq. 2.4. Substituting Eq. 2.3 into 2.4 yields an energy equation for a charged particle like the electron.

$$E = Fd \quad (2.4)$$

$$E = \mu_0 c^2 \left( \frac{\Delta e \Delta e}{4\pi r^2} \right) d \quad (2.5)$$

The distance ( $d$ ) at which force is exerted in this measurement is the electron's classical radius ( $r_e$ ), where  $r_e = 2.82 \times 10^{-15} \text{ m}$ . This is the sphere at which wave amplitude decreases at the square of the distance from the radius ( $r$ ), thus all are equal:  $d=r=r_e$ . Single charges are measured at this distance, thus there is no additional constructive wave interference and  $\Delta=1$ . These values are inserted into Eq. 2.5 and then simplified:

$$E = \mu_0 c^2 \left( \frac{e^2}{4\pi r_e^2} \right) r_e \quad (2.6)$$

$$E = \mu_0 c^2 \left( \frac{e^2}{4\pi r_e} \right) \quad (2.7)$$

The energy of two waves, into the electron and out of the electron, is measured as the energy of the electron. Eq. 2.7 is the rest energy of the electron. It is more commonly expressed as **electron mass ( $m_e$ )** as shown below.

$$m = \frac{E}{c^2} \quad (2.8)$$

$$m_e = \frac{\mu_0 e^2}{4\pi r_e} = 9.109 \cdot 10^{-31} \text{ [kg]} \quad (2.9)$$

The electron's mass can be derived from Ohm's law, with a known current and resistance of free space. In this respect, mass may be thought of as *resistance to change over time*.

### 3. @ Proton Radius

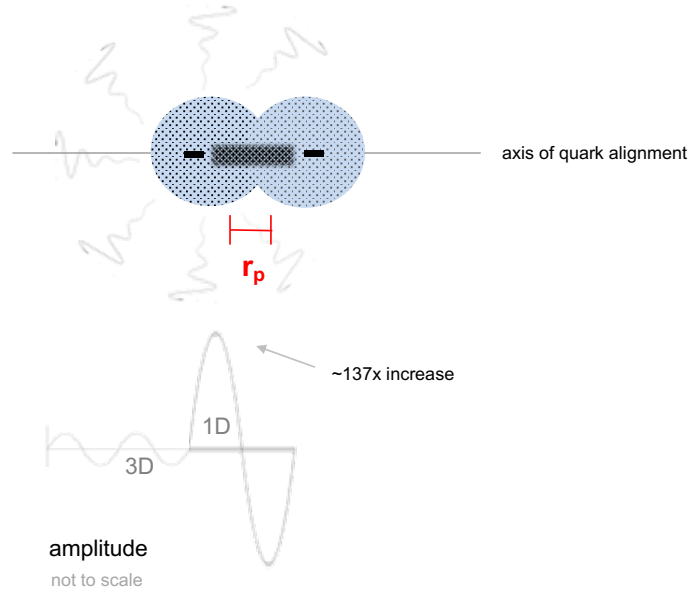
At distances around the size of the proton's radius or less, the strong force ( $F_s$ ) that binds quarks together to form protons and neutrons is known to be roughly 137 times stronger than the electromagnetic force ( $F_e$ ). Here, the symbol  $\Delta_s$  is temporarily used to demonstrate a change in wave interference, but it is more commonly known as the inverse of the fine structure constant ( $\alpha$ ).

$$F_s = \Delta_s F_e \quad (3.1)$$

$$\Delta_s = \frac{1}{\alpha} \quad (3.2)$$

The binding force that holds quarks together is very strong, but it is limited to short distances. In the next illustration, imagine two elementary charges separated at a distance of the proton's radius. Although the actual distance between quarks is not known, it is assumed to be less than the proton's radius since quarks form the proton.

In the illustration, two particles of the same charge are locked at a certain distance from each other, potentially at a stable distance if they reside at the node of a standing wave where there is no wave amplitude. The wave interference changes from a wave that expands spherically, to one that is concentrated in the axial direction between the two particles, forming what is perceived to be a gluon between quarks.



**Figure 3** – Two elementary charges separated a distance of the proton’s radius ( $r_p$ ). Distant charges continue to flow into the particle from three dimensions (3D). The charge is transformed in the direction of the second particle (1D) increasing wave amplitude only along the axis between the two charges.

The electric force (from Eq. 2.3) is now renamed  $F_e$ , to keep it distinct from the strong force  $F_s$ . Substituting it into Eq. 3.1 yields Eq. 3.4, which can be expressed as two elementary charges where one charge has an increase in wave amplitude before reaching the other charge as a result of the wave transformation described in the illustration above.

$$F_e = \mu_0 c^2 \left( \frac{\Delta e \Delta e}{4\pi r^2} \right) \quad (3.3)$$

$$F_s = \mu_0 c^2 \left( \frac{\Delta_s e e}{4\pi r^2} \right) \quad (3.4)$$

Next, the force is measured for two elementary particles,  $\Delta=1$ , at the proton’s radius ( $r_p$ ), where  $r_p = 8.663 \times 10^{-16} \text{ m}$ . The fine structure constant ( $\alpha$ ) replaces  $\Delta_s$  (from Eq. 3.2). The *strong* force at this distance is 42,127 N, which is ~137 times greater than the *electric* force at the same distance.

$$F_p = \mu_0 c^2 \left( \frac{e^2}{4\pi r_p^2} \right) \left( \frac{1}{\alpha} \right) = 42127 \left[ \frac{\text{kg} * \text{m}}{\text{s}^2} \right] \quad (3.5)$$

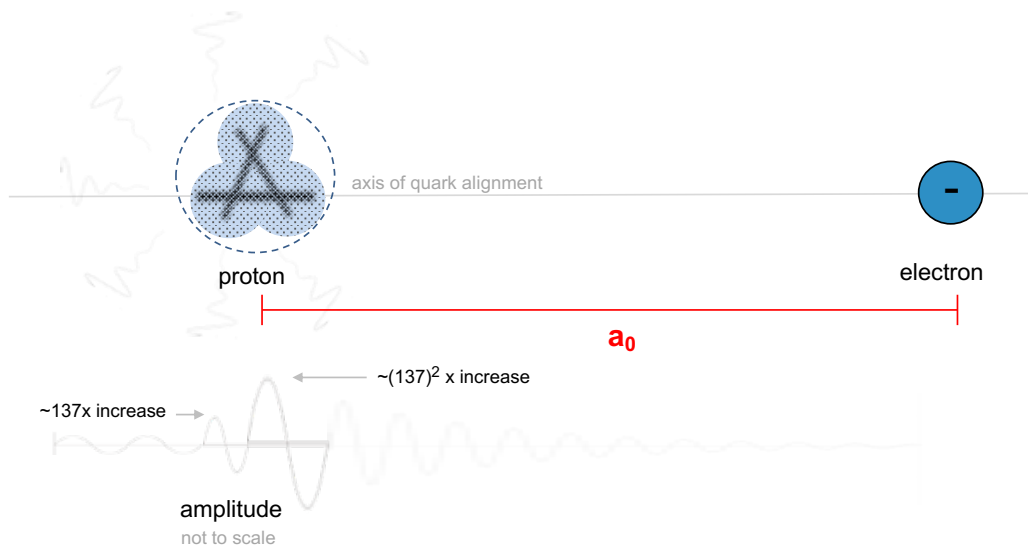
In this section, the calculation of the strong force is derived from what is already known as the relationship between the strong force and the electromagnetic force, but it yields the true form of the equation. The real proof of the fine structure constant and the wave transformation within a nucleon is detailed in the next section.



#### 4. @ Bohr Radius

The most common atomic element in the universe is hydrogen, where a single positive charge (proton) separates a single negative charge (electron) at the most *probable* distance known as the Bohr radius. The charges at this distance can be measured with the electric force established in this paper (Eq. 3.3). It is well known that the electron is attracted to the proton in the atom, yet it does not collapse into the nucleus. Thus, the electron is seen to orbit the nucleus in some fashion, yet to be determined. The model herein proposes that there is another force that is acting on the electron, which prevents it from spiraling into the nucleus of the atom. This previously unknown repulsive force is given the symbol “o” for orbital force ( $F_o$ ).

Expanding on the description of the strong force, when the wave interference of the proton changes from a spherical wave to a concentrated wave in the axial direction between two quarks, a wave transition occurs in both quarks before reaching the electron in its orbit. This only occurs on the axis between two quarks, creating a repulsive force for the electron when it intersects with the axis line. Thus, the electron experiences the normal attractive force with the proton while in orbit with the exception of when it intersects with a line where two or more quarks are in alignment.



**Figure 4** – Proton and electron separated at the Bohr radius ( $a_0$ ), the most probable location of the electron in the hydrogen atom when on the axis aligned with the quarks of a proton. Waves in the direction of the electron go through two quarks, increasing amplitude twice in the process (the square of the inverse fine structure constant -  $\sim 137^2$ ).

Consistent with the wave change that results in the strong force, there is a second wave interference pattern for the second quark. The strong force (from Eq. 3.4) is now expanded to include a second  $\Delta_s$ . Its transformation is impacted by the wave interference from the nearby electron, at a ratio of the electron radius and spherical radius ( $r_e/r$ ), creating a force that decreases with the **cube of distance**. This is consistent with the magnetic force from a static dipole source. In Eq. 4.2, the fine structure constant ( $\alpha$ ) replaces the change in wave interference ( $\Delta_s$ ) and the equation is simplified.

$$F_o = \mu_0 c^2 \left( \frac{\Delta_s e \Delta_s e}{4\pi r^2} \right) \left( \frac{r_e}{r} \right) \quad (4.1)$$

$$F_o = \mu_0 c^2 \left( \frac{e^2 r_e}{4\pi r^3 \alpha^2} \right) \quad (4.2)$$

The force along the axis between two quarks decreases with the **cube of distance**. This force ( $F_o$ ) repels the electron, but it decreases in intensity quickly. Meanwhile, the electron is also attracted to the proton due to the electric force, which decreases in intensity with the **square of distance**. Put simply, the orbital force ( $F_o$ ) prevents the electron from falling into the nucleus, and the electric force ( $F_e$ ) prevents the electron from escaping the atom. In this setup, there is a sweet spot where the repulsive orbital force ( $F_o$ ) and the attractive electric force ( $F_e$ ) cancel. Setting Eqs. 4.2. and 3.3 to be equal leads to the following:

$$F_o = F_e = \mu_0 c^2 \left( \frac{e^2 r_e}{4\pi r^3 \alpha^2} \right) = \mu_0 c^2 \left( \frac{e^2}{4\pi r^2} \right) \quad (4.3)$$

Cancelling common terms and then solving for radius ( $r$ ) in Eq. 4.3, gives the **Bohr radius ( $a_0$ )**.

$$\frac{r_e}{r^3 \alpha^2} = \frac{1}{r^2} \quad (4.4)$$

$$a_0 = r = \frac{r_e}{\alpha^2} = 5.292 \times 10^{-11} \text{ [m]} \quad (4.5)$$

Setting these two equations to be equal explains why we need the fine structure constant, which is the ratio of the strong force to the electric force, in an equation for orbitals. It also explains the Bohr radius. The *dimensionless fine structure constant ( $\alpha$ )* can be calculated from Eq. 4.5 as:

$$\alpha = \sqrt{\frac{r_e}{a_0}} = 0.007297 \quad (4.6)$$

The orbital force ( $F_o$ ) creates a binding energy between electrons and the atomic nucleus, when exerted over a distance ( $d$ ). Eq. 4.3 is then substituted into the following equation (4.7):

$$E_o = F_o (d) \quad (4.7)$$

$$E_o = \mu_0 c^2 \left( \frac{e^2 r_e}{4\pi r^3 \alpha^2} \right) (d) \quad (4.8)$$

The axial line between quarks extends to both sides of the atomic nucleus, such that there is a position for an electron on each side of the nucleus. The distance ( $d$ ) where energy is applied is the same as the spherical radius distance ( $r$ ), so when measured at the Bohr radius, it is  **$d=r=a_0$** . When half (**1/2**) of the energy of  $E_o$  is considered at this distance, it becomes the binding energy for the electron on one side of the nucleus. The binding energy for hydrogen is the **Rydberg unit of Energy ( $R_y$ )**, referred to here as an energy term ( $E_{Ry}$ ).

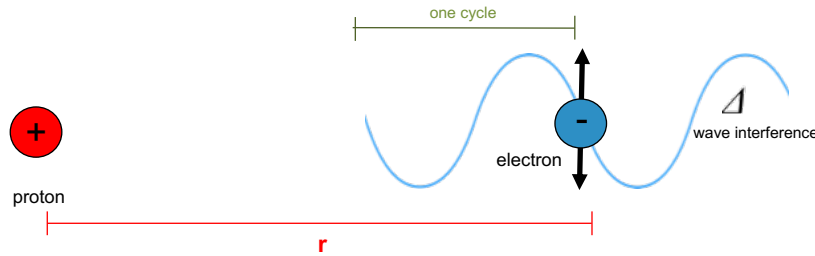
$$E_{Ry} = \frac{1}{2}E_o = \frac{1}{2}\mu_0 c^2 \left( \frac{e^2 r_e}{4\pi a_0^2} \right) = 2.18 \times 10^{-18} \left[ \frac{kg * m^2}{s^2} \right] \quad (4.9)$$

When this energy is released, it takes the form of a transverse wave, which is seen as the photon. The photon's energy is directly proportional to its frequency (f), as expressed in the Planck relation  $E=hf$ , where h is the Planck constant. Frequency is the inverse of time (t), represented by the time that it takes for a wave traveling at the speed of light (c) to travel a distance (r) and back, which is a cycle. A cycle is therefore two times the distance, or 2r. Frequency is the total number of cycles ( $n_{cycles}$ ) in this time period.

$$t = \frac{2r}{c} \quad (4.10)$$

$$f = \frac{n_{cycles}}{t} \quad (4.11)$$

Determining the number of cycles is not straightforward because there is an issue buried in the Planck relation. The photon's frequency is not solely based on distance (r), which is found as the only variable for time in Eq. 4.10. If it was solely based on this distance, it would be simpler to find a relationship between photon energy and atomic element configurations. For hydrogen, the energy relation is complex and results in different equations, such as the Lyman, Balmer and Paschen series, that describe an electron in different orbitals of quantum numbers. For other elements, it is more complex and no such equations exist to calculate photon energy. Thus, frequency is hiding another variable along with distance (r). The second is the variation of wave interference, which is based on the number and configuration of protons in the atomic nucleus and other electrons in the atom. This is why it is difficult to predict photon energies beyond a single electron in orbit. If energy continues to be related to the Planck constant (h), then frequency needs to be expanded by its true units, which is both the number of cycles and the distance at which time is determined. The number of cycles is described in Figure 5, which now includes the variables for both distance (r) and wave interference ( $\Delta$ ), including constants.



**Figure 5** – Transverse wave creation from proton and electron interaction.

A vibrating particle may create a transverse wave. The vibration from one point of equilibrium and back is  $1\pi$  in a sine wave, and the motion in the other direction and back to equilibrium completes one cycle ( $2\pi$ ). The speed of the vibration is determined by the wave interference. The reference point is a single proton and electron, in which the fine structure constant ( $\alpha$ ) appears again as the wave interference from the proton. A change ( $\Delta$ ) in wave interference as a result of other particles in the vicinity causes a change in vibration speed, thus changing the number of cycles. This is expressed mathematically as:

$$n_{cycles} = \Delta \frac{\alpha}{2\pi} \quad (4.12)$$

Inserting the number of cycles into Eq. 4.11, yields the following Eq. 4.14.

$$f = \Delta \frac{\alpha}{2\pi} \left( \frac{c}{2r} \right) \quad (4.13)$$

$$f = \Delta \frac{\alpha c}{4\pi r} \quad (4.14)$$

This equation can be tested on a single proton and electron (hydrogen) at the Bohr radius, where  $\Delta=1$  and  $r=a_0$ . The frequency matches the **photon for hydrogen ionization** for the first orbital. Frequencies for elements from hydrogen to calcium are calculated with a similar methodology and are presented in a separate paper [12].

$$f_0 = \frac{\alpha c}{4\pi a_0} = 3.29 \times 10^{15} \left[ \frac{1}{s} \right] \quad (4.15)$$

Returning to the Rydberg unit of energy (from Eq. 4.9), the variables for wave interference ( $\Delta$ ) and distance ( $r$ ) are inserted back into the equation to solve for atoms beyond hydrogen. The frame of reference for the change in wave interference was set to the Bohr radius ( $a_0$ ), so it remains in the equation as one distance ( $r$ ).

$$E = \frac{1}{2} \mu_0 c^2 \left( \frac{\Delta e^2 r_e}{4\pi r \alpha^2} \right) \left( \frac{1}{a_0} \right) \quad (4.16)$$

The constants can be rearranged as follows in Eq. 4.17. The squares of the speed of light and fine structure constants are separated to both sides of parentheses terms (outside and inside of the parentheses).

$$E = \frac{1}{2} \mu_0 c \frac{e^2}{\alpha} \left( \Delta \frac{r_e c}{4\pi r \alpha a_0} \right) \quad (4.17)$$

The terms on the right in parentheses can be further simplified. From Eq. 4.5, the Bohr radius can be replaced, resulting in Eq. 4.18. The terms on the right in parentheses is frequency from Eq. 4.14. The terms on the left (outside of parentheses) are the **Planck constant**.

$$E = \frac{1}{2} \mu_0 c \frac{e^2}{\alpha} \left( \Delta \frac{\alpha c}{4\pi r} \right) \quad (4.18)$$

$$E = h(f) \quad (4.19)$$

$$h = \frac{1}{2} \mu_0 c \frac{e^2}{\alpha} = 6.626 \cdot 10^{-34} \left[ \frac{kg \cdot m^2}{s} \right] \quad (4.20)$$

The photon's energy is a complex calculation of distance and wave interference from multiple particles. Yet, when each variable is understood, it can be derived from a single equation.

## Conclusion

The physics of electronic circuits is shown to be identical to the physics of subatomic particles and light. Although it is not surprising, since electricity is the motion of particles like the electron, it has been impossible to reconcile equations like Ohm's law with equations that describe matter, primarily because of the units used to describe components of these equations.

When the unit of charge is revisited, and it is measured as a distance and explained as wave amplitude that experiences constructive wave interference, then equations can be reconciled across the domains of quantum mechanics, classical mechanics and electronics. One such reconciliation presented in this paper is voltage, now represented as a force in classical units of the newton [ $\text{kg}\cdot\text{m}/\text{s}^2$ ].

Electronic concepts were then applied to subatomic particles in this paper, taking voltage measurements across three distances: 1) electron's classical radius, 2) proton's radius and 3) the Bohr radius (distance between the electron and proton in a hydrogen atom). With the correct units to express voltage, current and resistance, it is found that many of the fundamental physical constants that represent particles and their interactions can be derived directly from Ohm's law. One may conclude from this derivation that the forces acting upon the particles in our universe is based on an electric universe, in which electricity is not solely a force, but it is directly responsible for the electron's mass and the properties of light.

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